MILITARY SPECIFICATION

CANNON, 40MM DUAL AUTOMATIC GUN: M2A1

AND CLIP, CHARGER: M3

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification supports the acquisition of the recoil-operated, dual 40MM M2A1 cannon for single and automatic firing and the M3 charger-clip for feeding the cartridges.

1.2 This specification includes the minimum essential technical and Engineering Requirements and the necessary Quality Assurance Provisions to determine that all requirements for the purchased cannon have been met.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified in the solicitation, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-CCB-SAS, Watervliet, NY 12189-4050 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSMC N/A FSC 1010

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SPECIFICATIONS

Military

MIL-C-13931 - Cannon, General Specification for
MIL-P-14232 - Parts, Tools and Equipment for Army Material, Packaging of
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1189 - Standard Department of Defense Bar Code Symbology

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified in the solicitation, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. Army Armament, Munitions and Chemical Command

Gage 5276650	- Gage, Firing Pin Protrusion
5583768	- Clip, Charger, M3
EPL 5583768	- Engineering Parts List for Clip, Charger, M3
Gage, 7247039	- Gage, Headspace
Gage, 7247040	- Gage, Headspace
7305371	- Gun, Automatic 40MM M2A1, General Arrangement
EPL 7305371	- Engineering Parts List for Cannon, 40MM Dual Automatic Gun: M2A1
Master LQAP 5583768	- Master List of Quality Assurance Provisions for Clip, Charger, M3
Master LQAP 7305371	- Master List of Quality Assurance Provisions for Cannon, 40MM Dual Automatic Gun: M2A1
PDS P7305371	- Packaging Data Sheet, Gun, Automatic 40MM, M2A1
QAP-APPENDIX-WVA	- General Quality Assurance Provisions

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection iwth specific acquisition functions should be obtained from the contracting activity or as directed by the Procuring Contracting Officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified in the solicitation, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified in the solicitation, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM

ASTM D3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the AMerican Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. The applicable contract contains an order precedence clause wherein the specifications are listed last. With respect to this specification, the order of precedence, is drawing requirements, followed by the text of this specification, then the other referenced specifications and standards. Nothing in this specification shall supercede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Cannon and component parts shall be in accordance with the applicable contract, the General Arrangement 40MM Dual Automatic M2A1 Cannon drawing 7305371, the drawings and specifications listed in the Engineering Parts List EPL 7305371. Clips and component parts shall conform to the M3 Clip, Charger drawing 5583768, the drawings and specifications listed in the Engineering Parts List EPL 5583768. Cannon and clips shall also conform to the requirements of this specification, the provisions of MIL-C-13931, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.3).

3.2 First article. The contractor shall submit a first article unless it is specifically waived in the contract (see 6.2). No first article requirements shall be waived without review and approval by the Procuring Contracting Officer of the design activity.

3.3 Operation.

3.3.1 Cannon. The cannon shall operate within the designed rate of fire without interference, erratic movement or malfunction.

3.3.2 Charger-clips. The assembled clips shall hold each round firmly in proper position. Each hook of the assembled clips shall move to extended position under applied pressure and return to retracted position under spring pressure without lag, sticking or binding.

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3.4 Mechanical requirements.

3.4.1 Breech casing bodies. Preparatory to machining the fitted bolt holes, the breech casings shall be so assembled in matched pairs as to insure correct alignment of the trunnions and the recoil guideways.

3.4.2 Firing pins. The point of the firing pins may be finished with crocus cloth but shall not be hand stoned. The protrusion of the firing pin from the face of the breechblock shall be not less than 0.099 inch nor greater than 0.114 inch.

3.4.3 Headspace. Headspace shall be not less than 0.002 inch nor greater than 0.024 inch. (Headspace is the clearance between the forward face of the breechblock and the base of a maximum cartridge when both are in the firing position.)

3.5 Recoil cylinders. The recoil mechanism shall be filled with the type and volume of fluid specified on the recoil cylinder assembly drawing.

3.6 Proof testing. Cannon, replacement tubes, and replacement flash suppressors shall be subject to proof testing.

3.7 Performance.

3.7.1 Firing. Cannon and charger clips shall function properly during single and automatic firing. Irregularities and defects shall be corrected before acceptance.

3.7.2 High pressure resistance. The cannon shall be capable of withstanding a chamber pressure of 48,600 plus or minus 1,700 pounds per square inch when tested as specified in 4.6.2.3.

3.7.3 Firing rate. The firing rate on AUTO-FIRE shall be not less than 120 rounds per minute from each of the cannon, at approximately 70F ambient temperature.

3.7.4 Recoil. The recoil cylinder of each cannon shall be adjusted so that the recoil shall be not less than 7.4 inches nor more than 8.3 inches.

3.7.5 Material soundness. After proof firing, all parts shall be free of cracks and fractures and shall comply with the soundness requirements specified on the applicable documents.

3.7.6 Bore enlargementproof firing no more than 15 rounds, shall not exceed the inside (minimum bore) diameter specified on the applicable drawing by more than the enlargement specified in TABLE I.

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TABLE I. Bore enlargement.

Distance forward of commencement of rifling	0.10 inch	1 cal.	2 Cal. to muzzle
Enlargement (inch)	0.004	0.004	0.003

NOTE: Interpolate linearly to determine the allowable bore enlargement at distance other than the distances given.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all of the requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article (see 4.3).
b. First article initial production test, FA-IPT (see 4.4).
c. Quality conformance (see 4.5).

4.3 First article inspection. Unless otherwise specified in the contract, (see 6.2), First Article cannon and specified parts thereof shall be inspected by the contractor and submitted to the Government for approval. Inspection shall be in accordance with applicable Quality Assurance Provisions (QAPs) and QAP-APPENDIX-WVA. QAPs are listed in Master LQAP 7305371. First article sample sizes shall be as specified in applicable item QAPs, or the contract, as applicable. First article cannon and parts therof shall be representative of the production processes to be used during quantity production.

4.4 First Article - Initial Production Test (FA-IPT). When specified in the contract, (see 6.2), First Article cannon shall be subjected to Initial Production Testing by the Government or any agency specified by the Government and may be conducted at a Government facility to assure compliance with the requirements of

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this specification and any additional requirements listed in the contract. Initial Production Testing procedures and the sample size of First Article cannon shall be as specified in the contract. First Article cannon and parts thereof shall be representative of the production processes to be used during quantity production.

4.5 Quality conformance inspection.

4.5.1 Examinations.

4.5.1.1 Inspection provisions. Cannon and charger-clips and parts thereof shall be inspected for conformance to requirements of this specification, MIL-C-13931, QAPs listed in Master LQAP 7305371 and Master LQAP 5583768, and QAP-APPENDIX-WVA.

4.5.1.2 Breech casing bodies. Alignment of the trunnions and the recoil guideways shall be gaged to assure compliance with 3.4.1.

4.5.1.3 Firing pin protrusion. Protrusion of the firing pin from the face of the breechblock shall be measured with firing pin protrusion gage 5276650 or an approved equivalent, to determine compliance with the requirements of 3.4.2.

4.5.1.4 Headspace. Headspace shall be measured with headspace gages 7247039 and 7247040 or an approved equivalent to determine compliance with the requirement of 3.4.3.

4.5.1.5 Recoil cylinders. The recoil mechanism shall be examined to assure it is filled with the proper type and volume of fluid as specified in 3.5.

4.5.1.6 Final examination. Final examination of each cannon and each charger-clip shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract.

4.5.1.7 Inspection of packaging. Inspection of the Packaging requirements of Section 5 shall be performed to assure compliance with Section 5 and MIL-P-14232. Non compliance shall be cause for rejection.

4.5.1.8 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.5.2 Tests.

4.5.2.1 Functioning of cannons. Each cannon shall be tested to assure smoothness of operation, and absence of interference, binding, erratic movement and malfunction. Failure to comply with the requirements of 3.3.1 shall be cause for rejection.

4.5.2.2 Functioning of charger-clips. Charger-clips shall be tested with dummy rounds to assure compliance with the functional requirements of 3.3.2.

4.5.2.3 Proof testing.

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4.5.2.3.1 Cannon. Unless otherwise specified in the contract, each cannon with flash suppressor shall be proof fired by Government personnel at a proving ground designated by TECOM in accordance with TABLE II.

TABLE II. Cannon Firing Schedule.

No. of rounds	Pressure Limit (psi)	Elevation (Deg.)
1	48,600 ± 1,700	0
1	48,600 ± 1,700	75
8 (Auto-Fire)	43,000	75

4.5.2.3.2 Replacement tubes. Unless otherwise specified in the contract, each replacement tube shall be proof fired.

The firing schedule shall be in accordance with Table III.

4.5.2.3.3 Replacement flash suppressors. Unless otherwise specified in the contract, each replacement flash suppressor shall be proof fired. The firing schedule shall be in accordance with TABLE III.

Table III. Replacement tube and replacement flash suppressor firing schedule.

No. of Rounds	Pressure Limit (psi)
1	48,600 ± 1,700

As far as practical proof firing of replacement flash suppressors shall be combined with that of replacement tubes. If a combined test is impracticable, the flash suppressors shall be proof fired on a proof facility gun.

4.5.2.3.4 Performance. Cannon and charger clips shall be inspected to assure proper functioning during single and automatic firing to comply with 3.7.1.

4.5.2.3.5 Firing rate. The firing rate on AUTO-FIRE shall be determined to assure compliance with 3.7.3. The firing rate shall be determined at 75 degrees elevation at approximately 70F ambient temperature.

4.5.2.3.6 Recoil. The recoil shall be measured to determine compliance with 3.7.4.

4.5.2.3.7 Material soundness. After proof firing, the flash suppressor, breech ring, breechblock, and the exterior surface of the tube shall be magnetic particle tested in accordance with applicable drawings and specifications. In addition, the bore shall be examined with a borescope.

4.5.2.3.8 Bore enlargement. The bore diameters across the lands of the cannon tubes shall be measured after proof firing no more than 15 rounds. Measurement shall be made with a star gage, an air gage, or other approved instrument to determine compliance with 3.7.6 and TABLE I.

4.5.2.3.9 Records. Cannon, replacement tubes, and replacement flash suppressors

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that pass the firing tests shall be furnished with applicable data on DA Form 2408-4 by the proving ground.

4.5.3 Inspection equipment.

4.5.3.1 Acquisition, maintenance, and disposition. Unless otherwise specified, responsibility for acquisition, maintenance, and disposition of inspection equipment prescribed in applicable item QAPs listed in Master LQAP 7305371 and Master LQAP 5583768, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.3.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

5. PACKAGING

5.1 Cannon - Levels A, B and C. Preservation, packaging and packing of the cannon shall be in accordance with Packaging Data Sheet P7305371 for Gun, Automatic 40MM, M2A1 for the level of protection specified in the contract (see 6.2.1).

5.2 Replacement parts.

5.2.1 Levels A, B, and C. Replacement parts preservation, packaging and packing shall be in accordance with the applicable packaging data sheet for the level of protection specified in the contract (see 6.2.1).

5.2.2 Level X-commercial packaging. Replacement parts preservation, packaging, and packing shall be in accordance with ASTM D 3951 when Level X protection is specified in the contract (see 6.2.1).

5.3 Marking.

5.3.1 Levels A, B and C. Marking of unit packages and exterior shipping containers shall be in accordance with Packaging Data Sheet P7305371 for Gun, Automatic 40MM, M2A1 for the level of protection specified in the contract (see 6.2.1).

5.3.2 Level X - Commercial. Marking shall be in accordance with ASTM D 3951.

5.3.3 Bar code marking. When specified, (see 6.2.1) bar code marking shall be in accordance MIL-STD-129 and MIL-STD-1189.

6. NOTES

6.1 Intended use. The Gun, Automatic 40MM, M2A1 is intended for use in the Gun, Antiaircraft, Self-Propelled, Twin, 40MM, M42A1 System.

6.2 Ordering data. Procurement documents should specify the following:

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6.2.1 Acquisition requirements.

a. Title, number and date of this specification.
b. Requirements for and the number of cannon, components thereof, or both to be submitted for first article by the contractor (see 3.2).
c. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.
d. All inspection and test data that is required.
e. At whose expense a retest may be performed.
f. Serial numbers for the cannon and replacement tubes; flash suppressors and replacement flash suppressors.
g. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.3).
h. Availability of inspection equipment from the Government.
i. Extent of supplier's responsibility for Government-furnished and for supplier-required final inspection equipment.
j. The levels of preservation, packaging, and packing required for cannon assemblies (see 5.1) and replacement parts (see 5.2).
k. When applicable, bar code marking requirements (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.2	First Article Inspection Report	DI-T-4902

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Volume II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Contracting Officer).

6.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate.

6.4 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-STD-45662.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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U.S. GOVERNMENT PRINTING OFFICE: 1986 - 605-036/43954

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-14385C	CANNON, 40MM DUAL AUTOMATIC GUN: M2A1 AND CLIP, CHARGER: M3

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION (YYMMDD)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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